

2D Vector Operations

Symbolic Expressions for $\mathbf{A} = [a_1, a_2]$ and $\mathbf{B} = [b_1, b_2]$

1. Norm (Magnitude) of a Vector

The norm represents the length of the vector from the origin.

$$||\mathbf{A}|| = \sqrt{a_1^2 + a_2^2}$$

2. Distance Between Two Vectors

The straight-line Euclidean distance between the two points in a 2D space.

$$||\mathbf{A} - \mathbf{B}|| = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

3. Dot Product

The sum of the products of their corresponding components.

$$\mathbf{A} \cdot \mathbf{B} = a_1b_1 + a_2b_2$$

4. Cosine Similarity

Measures the angle (direction) between the vectors, independent of magnitude.

$$\text{Cosine Similarity}(\mathbf{A}, \mathbf{B}) = \frac{\mathbf{A} \cdot \mathbf{B}}{||\mathbf{A}|| \cdot ||\mathbf{B}||} = \frac{a_1b_1 + a_2b_2}{\sqrt{a_1^2 + a_2^2} \cdot \sqrt{b_1^2 + b_2^2}}$$

5. Cosine Distance

Measures the angular distance between the vectors.

$$\text{Cosine Distance}(\mathbf{A}, \mathbf{B}) = 1 - \text{Cosine Similarity}(\mathbf{A}, \mathbf{B})$$